# JUAN SEBASTIAN PEÑA DONNEYS

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## SUMMARY

Software engineer who builds and ships AI agents end-to-end, from the model to the customer. Sole engineer on a production LLM agent that quotes and charges real customers, builder of a voice agent that negotiates freight rates, and technical founder of an anti-fraud ticketing platform. I keep LLMs inside guardrails written in code and prove it with measurement: tests, adversarial evaluations and latency in live voice sessions.

## EXPERIENCE

**AI Software Engineer** — **Madrid · Remote**
**Strategos AI** — **Feb 2026 - Present**

- Sole engineer on Craig, a production AI agent for a Dublin print shop that quotes, charges and manages customer orders end-to-end over web chat and email.
- The LLM never invents a price: strict tool calling against a deterministic pricing engine (6 strategies) and about 15 server-side validation gates on every reply.
- Own the whole stack: agent design, FastAPI backend, Postgres data model, payments (Stripe), deployment on Cloud Run and operation. Report directly to the CEO and turn the non-technical founders' requirements into shipped features. In production since February 2026.

**Technical Founder** — **Colombia**
**Boletapp** — **2026 - Present**

- Anti-fraud ticketing platform (built, in testing, pre-launch): constantly rotating HMAC-derived QR codes and an offline-first door scanner that validates without connectivity.
- One cryptographic engine implemented twice, in TypeScript and Python, held bit-for-bit identical by shared test vectors.
- The problem it attacks: a ticket forwarded over WhatsApp stops working, because the code rotates on its own and opens the door exactly once.

**GenAI Mavericks - hackathon (1 week)** — **Madrid**
**Accenture** — **2025**

- First edition, 100+ students from 20 universities: a RAG solution turning legal documents into plain language (Justice 2030 Spain initiative).

## SELECTED PROJECTS

**Alex: voice AI freight negotiator, two articles in Towards AI**
*2026 · Python, LiveKit, GPT-4.1 mini, Next.js, GCP · talk-to-alex.web.app* — github.com/JSebastianIEU/voice-freight-negotiator

- Real-time voice agent that verifies the caller and negotiates freight rates in English and Spanish; a guided live voice demo in the browser.
- Price limits enforced in code, not in the prompt: across 10 adversarial attacks x 3 rounds, margin given away fell from $197 to $72 of $500 and limit leaks from 4/30 to 0/30.
- 1.39 s median response in live voice sessions; turn waiting cut from 3 s to 1.2 s and the LLM chosen by measured time to first token (705 ms vs 2,153 ms).
- 183 tests at 91% coverage, CI on every PR, automatic deploys to Cloud Run with Workload Identity Federation.

**TikTok Recommendation Platform**
*Team of 7 · Jan-Apr 2026 · Python, FastAPI, React* — github.com/PredictiveSocialMedia/Tik-Tok-Recommendation-System

- 98 of 282 commits, 22 pull requests opened and 9 of my teammates' PRs reviewed.
- Owned evaluation: NDCG@k ranking metrics, temporal splits respecting point-in-time boundaries, and 25 regression tests validating the metrics against analytically known ground truth.
- Built the video analysis pipeline (8 parallel branches: Whisper, OCR, captioning, keywords); cut it from over 5 minutes per video to under 60 seconds.

**Semantic Engagement on TikTok - published in Towards AI**
*2026 · Python, transformers, FAISS* — github.com/JSebastianIEU/tiktok-semantic-engagement

- Do semantically similar captions perform similarly? Across 22,647 posts, 12 Mann-Whitney U tests show 10.5-13% lower engagement variance than random baselines.

**QR Forge**
*Open source · 2026 · FastAPI, React, Docker, Azure* — github.com/JSebastianIEU/qr_forge

- 136 tests at 80% line coverage behind a CI gate; three-stage non-root Docker image; CI/CD to Azure with Trivy, Bandit and Safety scanning.

## EDUCATION

**IE University** — **Madrid, Spain**
**BSc in Computer Science and Artificial Intelligence** — **Expected: Jul 2027**

- CUBICO Young Talented Leaders Scholarship (academic and leadership excellence). Machine Learning: 9.5/10.

## TECHNICAL SKILLS

**Languages:** Python, TypeScript, JavaScript, SQL, Java, C

**AI and voice:** LLM agents and tool calling, guardrails and evaluation, RAG, embeddings, LiveKit, WebRTC, speech to text and text to speech (Deepgram, Cartesia), PyTorch, scikit-learn

**Backend and data:** FastAPI, Node.js, PostgreSQL, SQLModel, Alembic, pandas, REST APIs and webhooks
**Frontend:** React, Next.js, Tailwind

**Cloud and DevOps:** GCP (Cloud Run, Firebase), Azure, Docker, GitHub Actions, Prometheus, pytest, Vitest

## LANGUAGES

**Spanish** (native) · **English** (bilingual, daily professional use)